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RESEARCH ARTICLE
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Automated Estimation and Tools to Extract Positions, Velocities, Breaks, and Seasonal Terms From Daily GNSS Measurements: Illuminating Nonlinear Salton Trough Deformation

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Key Points:
- Positions, velocities, breaks, and seasonal terms for thousands of GNSS stations are updated every week
- These results can be used to study interseismic plate motion, coseismic deformation, postseismic deformation, and seasonal variations
- The use of these tools shows highly variable nonlinear motion of GPS stations in southern California

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Abstract This paper describes the methods used to estimate positions, velocities, breaks, and seasonal terms from daily Global Navigation Satellite System (GNSS) measurements. Break detection and outlier removal have been automated so that decades of daily measurements from thousands of stations can be processed in a few hours. New measurements are added, and parameters are updated every week. Model parameters allow separation of interseismic, annual, coseismic, and postseismic signals. Tools available through GeoGateway (http://geo-gateway.org) allow rapid visualization and analysis of these terms for results that can be subsetted in time or space. Results show highly variable and nonlinear motion for GPS stations in southern California. The variable motion is related to seasonal motions, distributed tectonic motion, earthquakes, and postseismic motions that can continue for years. In some areas results suggest that additional processes are responsible for the observed motions. In general, following earthquakes, stations return to their long-term motions after 2–3 years, though some exceptions occur. The use of the tools shows nonlinear motion in the Salton Trough of southern California related to the 2010 M7.2 El Mayor-Cucapah earthquake, 2012 Brawley earthquake swarm, and a creep event on the Superstition Hills fault in 2017.

1. Introduction

Global Navigation Satellite System (GNSS) measurements capture the shape of the solid earth and how it changes with time. Daily positions are estimated for thousands of receivers around the world. Analysis of GNSS data can be quite cumbersome requiring significant computing resources to process large data sets. In this paper we describe two layers of software designed to automate significant portions of the analysis and allow scientists to focus on deformation in their region of interest. The automated process generates both edited daily time series and a functional model for each receiver site. The model works well for most sites and allows prediction at any time. Time series data can be used directly for receiver sites with motion too complex to be captured by the model.

Two layers of software are utilized to deliver GNSS-derived parameters to GeoGateway (Donnellan et al., 2016) users. GeoGateway is a data product search and analysis gateway for scientific discovery, field use, and disaster response (http://geo-gateway.org). GeoGateway focuses on delivery of NASA geodetic imaging products from UAVSAR and GPS integrated with earthquake faults data sets, seismicity, and models. The first layer for delivering GNSS-derived parameters through GeoGateway is the Jet Propulsion Laboratory's (JPL) GipsyX software (Bertiger et al., 2020), which is used to compute edited time series, positions, velocities, breaks, and seasonal terms for thousands of GNSS sites every week (https://sideshow.jpl.nasa.gov/post/series.html). GipsyX has in-code documentation for those with a source license, command line help for those with an executable license, and on-line training materials with sample data sets for all users. We use GipsyX in precise point positioning (PPP) mode (Zumberge et al., 1997) with JPL 3.0 final orbits (Dietrich et al., 2018), applying IERS 2010 solid earth tides (Petit & Luzum, 2010), GYM95 satellite yaw models (Bertiger et al., 2020), GPT2w troposphere mapping functions and nominals (Böhm et al., 2015), and IGS antenna phase center maps. New measurements are added every week and all parameters are updated. The second layer is the web-based GeoGateway software (http://geo-gateway.org), which allows

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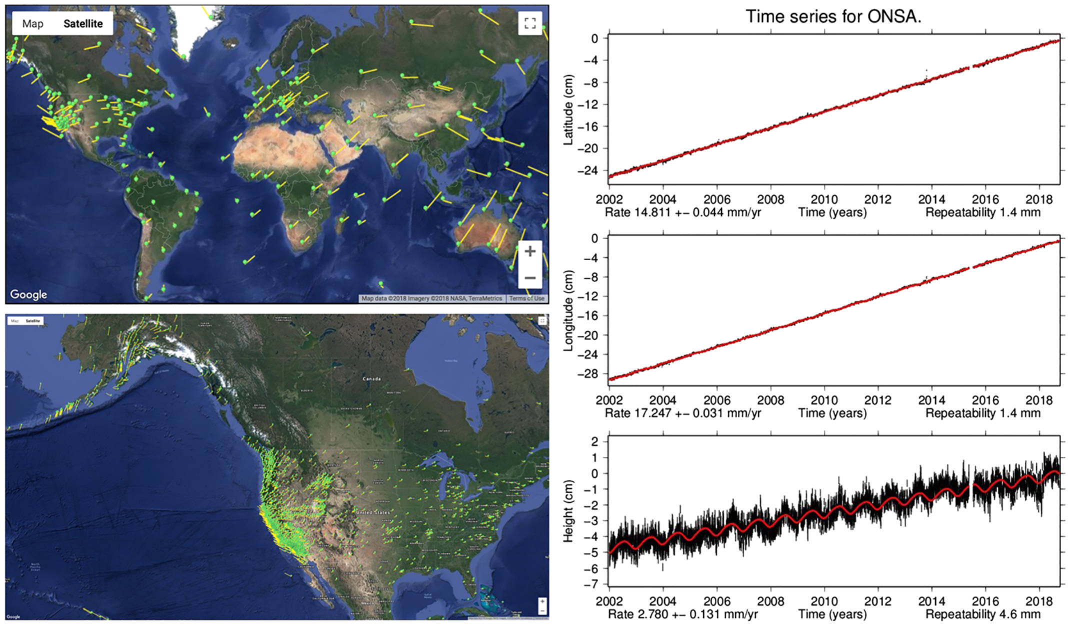

Figure 1. Global velocity field (top left) and North American velocity field (bottom left). Edited time series with error (black bars), model fit values (red line), velocity estimates, and repeatabilities for ONSA in Onsala, Sweden (right).

users to retrieve and manipulate GNSS products from JPL and integrate them with other data sources such as Interferometric Synthetic Aperture Radar (InSAR), seismicity, and mapped faults, and models. Both layers of software have been implemented using Python 3. The tools are also available as python scripts via Github (https://github.com/GeoGateway/GPS_Services) for command line use.

2. GipsyX Layer

2.1. Estimation

Each component of each site has one position, one velocity, an annual cosine term, an annual sine term, a semiannual cosine term, a semiannual sine term, and a variable number of breaks. Least-squares estimates are computed by fitting daily input measurements in three dimensions with full covariance information for all estimated site parameters based on the following equation of motion (Bock & Melgar, 2016; Dong et al., 2002):

$$X = x + v(t - \mathrm{t}_0) + a^*\cos(2\ \pi(t - t_0)) + b^*\sin(2\ \pi(t - t_0)) + c^*\cos(4\ \pi(t - t_0))$$
$$+\ d^*\sin(4\ \pi(t - t_0)) - B^*H\ (t_b - t),$$

where x is an initial position, v is a constant velocity, a is the annual cosine amplitude, b is the annual sine amplitude, c is the semiannual cosine amplitude, d is the semiannual sine amplitude, t is the current time in years, t_0 is the reference epoch for velocities and periodic terms, B is the size of the break which occurs at time t_b, and $H\ (t_b - t)$ is the Heaveside function that is zero for $(t_b - t) < 0$ and one for $(t_b - t) >= 0$.

Parameter estimation is carried out using the following iterative approach. A single file is created for each site containing all available position measurements and covariances. A square root information filter, which finds the least-squares solution, is used to compute a preliminary linear fit. A second linear fit is performed after coarse outlier removal. An exhaustive break search is carried out using an F test for significance. A linear fit with breaks is computed. A second linear fit with breaks is carried out after fine outlier removal. A second break search is run. A final linear fit with breaks and seasonal terms is computed.

2.2. Time Series

Edited time series, fit parameters, and residuals are provided for 2,674 globally distributed sites. Results for a representative site ONSA are shown in Figure 1. The black points with error bars are the edited GNSS

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measurements. The red points are computed from the fit parameters. Velocities and error bars are given for each component along with repeatability which is given as the wrms of residuals.

2.3. Velocity Field

Global and North American velocity fields are illustrated in Figure 1. Validation has been carried out by comparison with the IGS14 (Rebischung et al., 2016) velocity field on 1 January 2010. The weighted rms of position agreement is 1.2 mm N, 1.4 mm E, and 3.0 mm V. The weighted rms of velocity agreement is 0.2 mm/year N, 0.2 mm/year E, and 0.5 mm/year V.

2.4. Breaks

Various schemes for automated break detection in GPS time series have been investigated (Gazeaux et al., 2013). Here, every possible location in the time series is a candidate break. A preliminary fit is carried out with no breaks to establish the baseline value of χ^2. One break is added, and χ^2 is computed for every possible break location. An F test is computed for the break location that minimizes χ^2. If the F test is passed, that break is kept. An additional break is added, and χ^2 is computed for every possible remaining break location. An F test is again computed for the break location that minimizes χ^2. If the F test is passed, that break is also kept. This process is repeated until there are no additional breaks that pass the F test.

The F test equation is

$$F = \left[\left(\chi^2(wo) - \chi^2(w)\right)/\chi^2(w)\right] \times [(n\text{data} - pw)/(pw - pwo)],$$

where $n$data is the number of position observations being fit, pw is the number of parameters with the break included, pwo is the number of parameters without the break included, $\chi^2(wo)$ is the fit performed without a break (wo) and the sum of squared residuals computed, and $\chi^2(w)$ is the fit performed with a break and a new sum of squared residuals computed.

If the break is real then $\chi^2(w) < \chi^2(wo)$. A bigger reduction indicates that the break is more likely to be real. Changing the value of F adjusts the sensitivity of break detection. A value that is too small will result in too many breaks while a value that is too big will result in too few. Current products keep breaks when F is greater than 150. Each break adds three degrees of freedom and the probability of F exceeding 150 by chance is less than 1%. We expect no breaks for a subset of sites which are assumed to have stable linear motion, such as ONSA (Figure 1). Overall, 12% of sites have no breaks.

An empirical approach to break detection has been chosen to allow full automation of weekly updates. No knowledge from earthquake catalogs or site logs is used. Large breaks can be detected from a single point at the end of a time series. Smaller breaks are detected as additional measurements are added. If there is missing data during an event, the break time is assigned to the first measurement after the displacement has occurred The number of breaks gives a rough indication of how much complex motion exists at a particular site. Only 9% of sites have more than nine breaks.

2.5. Outliers

Outlier removal is based on either the size of the error bars for a particular measurement or on the number of σs for a particular residual. Coarse outlier removal is typically set to remove measurements with error bars larger than 10 cm and residual outliers greater than 10σ. Fine outlier removal is typically set to remove measurements with error bars larger than 5 cm. Once outliers are removed an edited time series is produced.

Outlier removal impacts break detection. If an earthquake occurs and a few points are added to the time series, these could easily be removed as outliers. In order to avoid this issue the current strategy keeps most points and only removes the worst 1% of daily measurements. Incomplete data from the receiver is one common cause of outliers.

2.6. Errors

Every parameter has an estimated formal error which is propagated through the entire analysis chain. These formal errors, however, are unrealistically small. In order to make them more realistic but still retain accurate relative information, a single scale factor is applied. The scale factor is determined using data decimation for a set of roughly 35 sites with stable velocities. A long span of data is broken into 3 years segments. Velocity estimates and unscaled formal errors are determined for each segment. The standard deviation of

Table 1
Velocity Values Shown in Figure 2 (Top Left Panel)

Station	Longitude	Latitude	E (mm/year)	N (mm/year)	V (mm/year)
Velocity					
COAG	−115.123348	32.863464	−13.4 ± 0.3	−7.7 ± 0.4	−2.1 ± 1.3
COKG	−115.728503	32.850513	−32.6 ± 0.3	11.4 ± 0.3	1.5 ± 1.0
CRRS	−115.735044	33.069808	−24.8 ± 0.2	7.2 ± 0.3	0.0 ± 0.9
ERRG	−115.822734	33.116446	−28.6 ± 0.3	5.2 ± 0.4	−0.9 ± 1.4
FSHB	−115.799944	32.944989	−32.0 ± 0.4	11.0 ± 0.4	0.9 ± 1.5
GLRS	−115.521374	33.274811	−17.1 ± 0.2	−8.6 ± 0.2	−0.6 ± 0.7
IID2	−115.031804	32.706168	−14.0 ± 0.1	−6.9 ± 0.2	−3.3 ± 0.6
IVCO	−115.506793	32.829383	−25.3 ± 0.5	10.5 ± 0.5	−2.1 ± 2.0
P492	−115.969454	32.887049	−35.6 ± 0.1	13.2 ± 0.1	0.5 ± 0.5
P493	−115.824894	32.954703	−33.0 ± 0.1	10.6 ± 0.1	1.2 ± 0.5
P494	−115.732069	32.759654	−33.3 ± 0.2	12.3 ± 0.3	1.6 ± 0.9
P495	−115.628395	33.044961	−21.9 ± 0.2	4.8 ± 0.2	−5.6 ± 0.8
P496	−115.595956	32.750626	−29.7 ± 0.2	9.4 ± 0.2	2.1 ± 0.7
P497	−115.577033	32.834735	−27.0 ± 0.2	9.0 ± 0.2	1.3 ± 0.7
P498	−115.569568	32.898431	−24.3 ± 0.3	7.4 ± 0.3	0.6 ± 1.1
P499	−115.487914	32.979606	−17.1 ± 0.2	−5.5 ± 0.3	−5.3 ± 0.9
P500	−115.299933	32.690046	−16.9 ± 0.2	−3.7 ± 0.2	−2.6 ± 0.7
P501	−115.397913	32.875756	−14.7 ± 0.1	−6.5 ± 0.2	−1.2 ± 0.5
P502	−115.421913	32.982451	−15.3 ± 0.1	−7.7 ± 0.2	−1.9 ± 0.5
P503	−115.720184	32.948915	−29.1 ± 0.2	8.5 ± 0.2	1.7 ± 0.8
P506	−115.510199	33.08143	−16.4 ± 0.4	−6.3 ± 0.5	−4.9 ± 1.7
P507	−115.612407	33.199975	−20.6 ± 0.1	−13.5 ± 0.2	−13.1 ± 0.5
P508	−115.428707	33.24778	−15.6 ± 0.2	−7.7 ± 0.2	−0.7 ± 0.7
P509	−115.293919	32.890662	−13.5 ± 0.2	−8.1 ± 0.3	−2.1 ± 0.9
P510	−115.343334	33.143574	−14.9 ± 0.1	−8.3 ± 0.2	−0.6 ± 0.6
P744	−115.508407	32.829382	−21.8 ± 0.2	9.0 ± 0.3	2.8 ± 1.0
PHJX	−115.550453	32.490056	−31.1 ± 0.1	14.8 ± 0.1	0.1 ± 0.5
PJZX	−115.883733	32.417008	−41.5 ± 0.2	17.8 ± 0.2	0.0 ± 0.7
PTAX	−115.456337	32.377487	−31.2 ± 0.3	16.5 ± 0.4	0.6 ± 1.2
SLMS	−115.977844	33.292228	−24.5 ± 0.1	4.4 ± 0.1	1.0 ± 0.5
WMDG	−115.581897	33.038324	−23.7 ± 0.5	−3.7 ± 0.6	−14.5 ± 2.1
WWFG	−115.57878	33.278835	−18.3 ± 0.3	−8.9 ± 0.3	−2.3 ± 1.1
YUHG	−115.922236	32.647566	−38.7 ± 0.4	16.2 ± 0.4	−0.4 ± 1.8

the velocity estimates over all segments is then divided by the mean velocity error for all segments to obaint the scaling factor which currently has a value of 20.

2.7. Computation

Weekly updates using the GipsyX layer are carried out on a cluster containing 32 Intel® Xeon® X5450 at 3.00 GHz processors with eight core nodes each. Processing can be done in parallel because point positions at different sites are independent. Weekly updates take less than 2 hr for 2,675 sites with 11.9 million station days of measurements from 1994 to 2019 including break detection, outlier removal, and least-squares estimation.

3. GeoGateway Layer

There are five main GeoGateway options for exploring the time series and parameter estimates. Four of them read fit parameters as input and one of them reads only the edited daily time series as input. All five tools output a table of values with error bars along with horizontal and vertical KML files which can be used to display results on a map. The velocity tool reads the estimates for each site in the region of interest. The coseismic tool adds all breaks in a time window around an event in order to compute coseismic displacements. A window is necessary in order to deal with data gaps and the fact that measurements are averaged over a full day. The default coseismic time window is ±0.05 years on either side of the event. The postseismic tool adds all breaks in a window following an event to compute postseismic displacements. The default post-seismic window starts 0.05 years after the event time and ends 2.05 years later. The model tool uses all fit

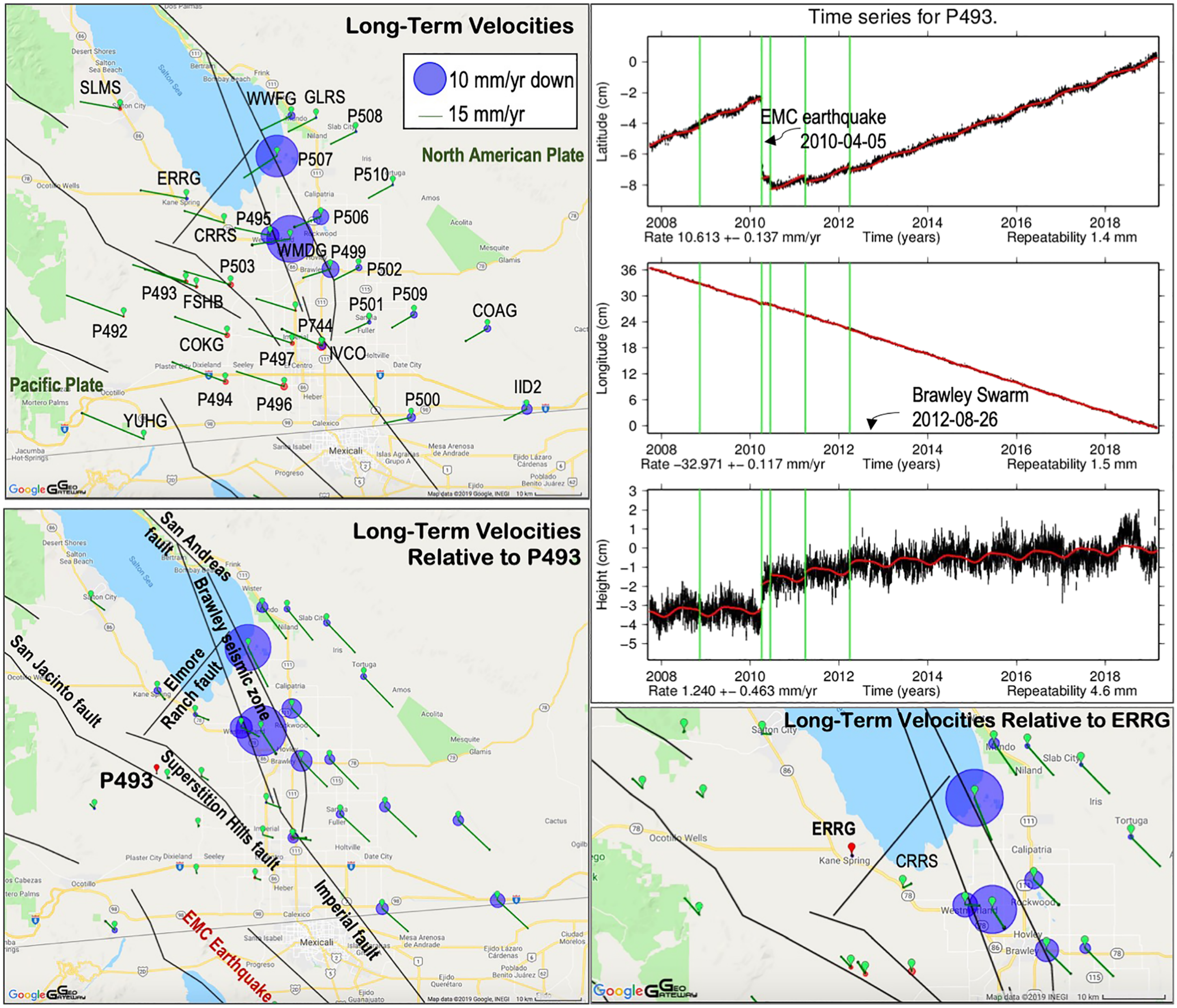

Figure 2. Velocity estimates for the Salton Trough in IGS14 (top left), velocities relative to station P493 (bottom left), velocities relative to station ERRG (bottom right), and time series for station P493 in IGS14 (top right). P493 is marked by a red icon in the lower left map. Velocity vectors extend from the station icons and include error ellipses, which are generally too small to see. Blue circles represent downward motion and red circles uplift. Large jump at station P493 shows is due to the 2010 M7.2 El Mayor-Cucapah (EMC) earthquake. Breaks in the time series are marked with vertical green lines. Positions with errors are marked by vertical black bars. The model fit is shown as a red line.

parameters to compute a displacement between two user specified times. The displacement tool avoids use of any fit parameters and instead uses daily time series data to directly compute displacements between any two times which actually have measurements. In addition to these five tools, users are provided with edited daily time series data so they can perform their own customized analysis if desired.

4. Results—Nonlinear Deformation of the Salton Trough

GeoGateway tools are useful for providing an intuitive and visual undersanding of crustal deformation data. GeoGateway allows for map-based presentation of multiple data sets, such as from InSAR/UAVSAR, seismicity, faults, user genreated KML files, and GNSS vectors presented in the various formats described above.

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Table 2
Velocity Values Relative to P493 Shown in Figure 2 (Bottom Left Panel)

Station	Longitude	Latitude	E (mm/year)	N (mm/year)	V (mm/year)
Velocity relative to P493					
COAG	−115.123348	32.863464	19.5 ± 0.3	−18.4 ± 0.4	−3.4 ± 1.3
COKG	−115.728503	32.850513	0.4 ± 0.3	0.7 ± 0.3	0.2 ± 1.0
CRRS	−115.735044	33.069808	8.2 ± 0.2	−3.4 ± 0.3	−1.3 ± 0.9
ERRG	−115.822734	33.116446	4.4 ± 0.3	−5.5 ± 0.4	−2.2 ± 1.4
FSHB	−115.799944	32.944989	0.9 ± 0.4	0.4 ± 0.4	−0.3 ± 1.5
GLRS	−115.521374	33.274811	15.8 ± 0.2	−19.2 ± 0.2	−1.8 ± 0.7
IID2	−115.031804	32.706168	18.9 ± 0.1	−17.6 ± 0.2	−4.6 ± 0.6
IVCO	−115.506793	32.829383	7.7 ± 0.5	−0.1 ± 0.5	−3.3 ± 2.0
P492	−115.969454	32.887049	−2.6 ± 0.1	2.6 ± 0.1	−0.7 ± 0.5
P493	*−115.824894*	*32.954703*	*0.0 ± 0.1*	*0.0 ± 0.1*	*0.0 ± 0.5*
P494	−115.732069	32.759654	−0.4 ± 0.2	1.7 ± 0.3	0.4 ± 0.9
P495	−115.628395	33.044961	11.1 ± 0.2	−5.8 ± 0.2	−6.8 ± 0.8
P496	−115.595956	32.750626	3.3 ± 0.2	−1.3 ± 0.2	0.9 ± 0.7
P497	−115.577033	32.834735	6.0 ± 0.2	−1.7 ± 0.2	0.0 ± 0.7
P498	−115.569568	32.898431	8.7 ± 0.3	−3.2 ± 0.3	−0.7 ± 1.1
P499	−115.487914	32.979606	15.8 ± 0.2	−16.1 ± 0.3	−6.6 ± 0.9
P500	−115.299933	32.690046	16.1 ± 0.2	−14.3 ± 0.2	−3.8 ± 0.7
P501	−115.397913	32.875756	18.2 ± 0.1	−17.1 ± 0.2	−2.4 ± 0.5
P502	−115.421913	32.982451	17.7 ± 0.1	−18.3 ± 0.2	−3.2 ± 0.5
P503	−115.720184	32.948915	3.9 ± 0.2	−2.1 ± 0.2	0.4 ± 0.8
P506	−115.510199	33.08143	16.5 ± 0.4	−16.9 ± 0.5	−6.1 ± 1.7
P507	−115.612407	33.199975	12.3 ± 0.1	−24.1 ± 0.2	−14.3 ± 0.5
P508	−115.428707	33.24778	17.4 ± 0.2	−18.3 ± 0.2	−2.0 ± 0.7
P509	−115.293919	32.890662	19.4 ± 0.2	−18.7 ± 0.3	−3.3 ± 0.9
P510	−115.343334	33.143574	18.1 ± 0.1	−18.9 ± 0.2	−1.8 ± 0.6
P744	−115.508407	32.829382	11.2 ± 0.2	−1.6 ± 0.3	1.5 ± 1.0
PHJX	−115.550453	32.490056	1.8 ± 0.1	4.2 ± 0.1	−1.2 ± 0.5
PJZX	−115.883733	32.417008	−8.5 ± 0.2	7.2 ± 0.2	−1.2 ± 0.7
PTAX	−115.456337	32.377487	1.8 ± 0.3	5.9 ± 0.4	−0.6 ± 1.2
SLMS	−115.977844	33.292228	8.5 ± 0.1	−6.2 ± 0.1	−0.2 ± 0.5
WMDG	−115.581897	33.038324	9.2 ± 0.5	−14.3 ± 0.6	−15.8 ± 2.1
WWFG	−115.57878	33.278835	14.7 ± 0.3	−19.5 ± 0.3	−3.5 ± 1.1
YUHG	−115.922236	32.647566	−5.7 ± 0.4	5.6 ± 0.4	−1.6 ± 1.8

Products generated through GeoGateway can be downloaded for further analysis and modeling. We present here an example of how GeoGateway GNSS analysis tools can be used to rapidly explore data for key features and further scientific analysis.

Long-term velocities in IGS14 (Table 1) show a change in the azimuth of velocity vectors across the Salton Trough (Figure 2). Long-term (decadal) velocities relative to station P493 (Table 2) qualitatively highlight the Salton Trough as the boundary between the Pacific and North American plates. Both IGS14 and plots relative to P493 show long-term subsidence of the Salton Trough. IGS14 velocities suggest weak uplift at the western margin of the Salton Trough near the Superstition Hills fault (SHF). Relative vertical velocities in the westward region are near zero, suggesting that area behaves as a block. For the remaining Figures 3 and 4 we show horizontal velocities relative to station P493 and vertical velocities in IGS14.

Time series for station P493 show complex motion (Figure 2). Five breaks have been determined for the site, which are represented by green vertical lines. The first may be due to a slight discontinuity in the vertical time series in November 2008. No significant earthquakes occurred in southern California in the months prior to then. The second break is due to the 4 April 2010 El Mayor-Cucapah (EMC) earthquake followed by a third break which fits a small amount of postseismic motion and two additional breaks, which cannot be explained by the interseismic rate. The station shows a relatively large amount of uplift at the time of the EMC earthquake with gradual continued uplift.

Coseismic displacements or breaks associated with the EMC earthquake have been reported using these tools in Donnellan et al. (2018). Here we show two interesting sets of breaks observed post EMC

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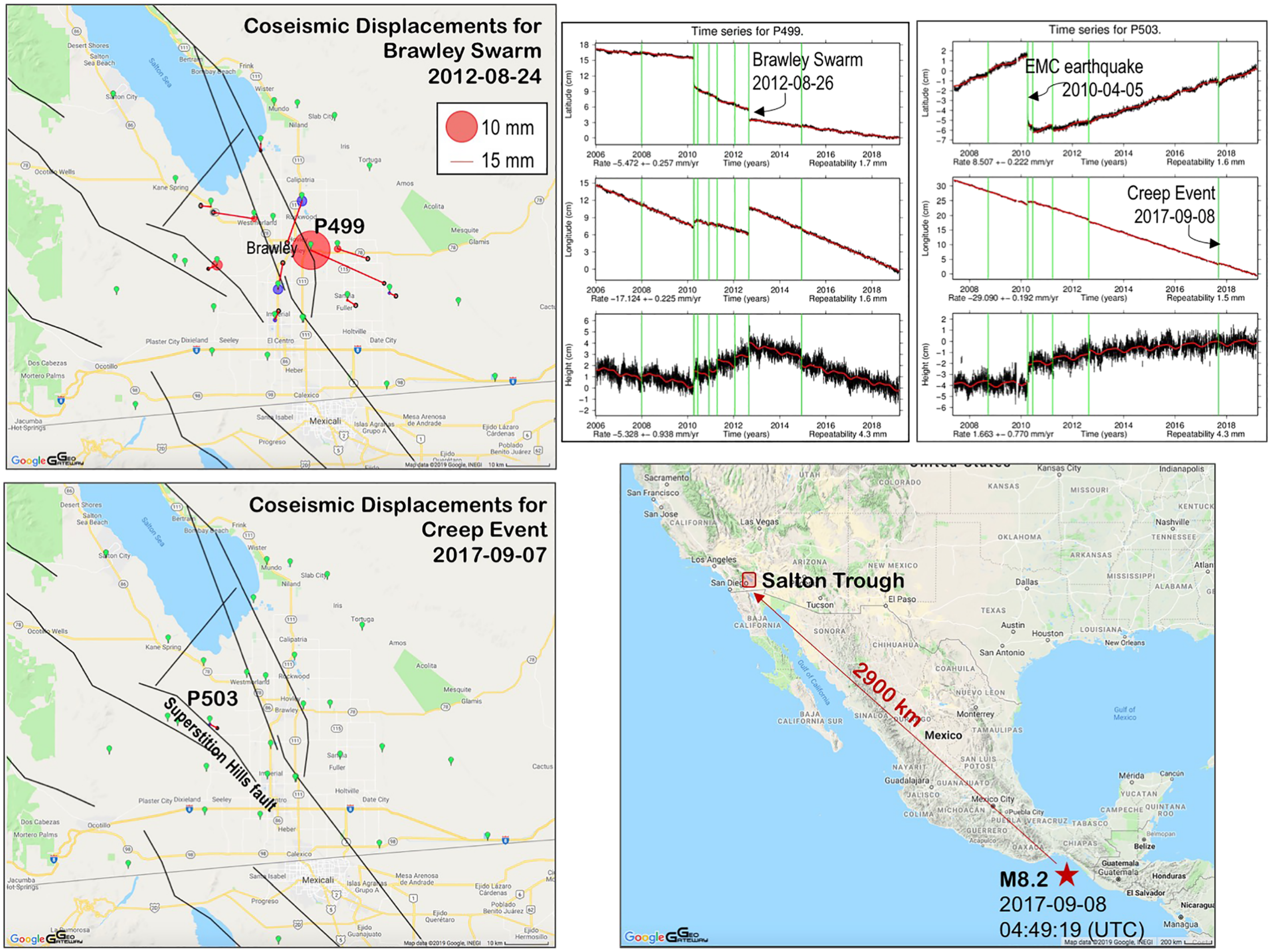

Figure 3. Coseismic displacements or automatically identified breaks for time surrounding the Brawley swarm that began 26 August 2012 (top left) and for a creep event on the Superstition Hills fault on 7 September 2017 (bottom left). A break was only detected at station P503 for the slip event. Time series for stations P499 and P503 (top right) show large jumps for the 2010 El Mayor-Cucapah (EMC) earthquake and the 2012 Brawley swarm. Additional breaks absorb motion outside the model. The 2017 break at P503 coincides in time with the M8.2 Chiapas earthquake (lower right).

earthquake (Figure 3). The Brawley earthquake swarm began on 26 August 2012 and included three moderate earthquakes of M_w5.3, M_w4.9, and M_w5.4 over a 90 min period (Hauksson et al., 2013). The swarm resulted in north-south shortening and east-west extension of the Salton Trough (Figure 3 and Table 3), which is consistent with northwest striking right-lateral and northeast striking left-lateral conjugate faulting. Results here are similar to those previously reported by Wei et al. (2013). Here we also show vertical breaks in the data, which highlight 12.8 ± 4.2 mm uplift of station P499 nearest to Brawley. Uplift occurred east and west of the swarm and a small amount of subsidence occurred north and south of the main swarm. Vertical time series are discussed below.

The swarm caused 6.4 ± 1.3 mm of westward motion of station P503 toward the Superstition Hills fault (Table 3). In September 2017 station P503 shows a second break in the time series. (Figure 3). Vector motion of 5.4 ± 1.3 mm of the station is nearly parallel to the SHF. The magnitude of slip and sense of motion of this break is comparable to the long-term rate of station P503 of 4.4 ± .3 mm/year to the southeast. There is no appreciable vertical motion either long term or episodically for this slip event at station P503.

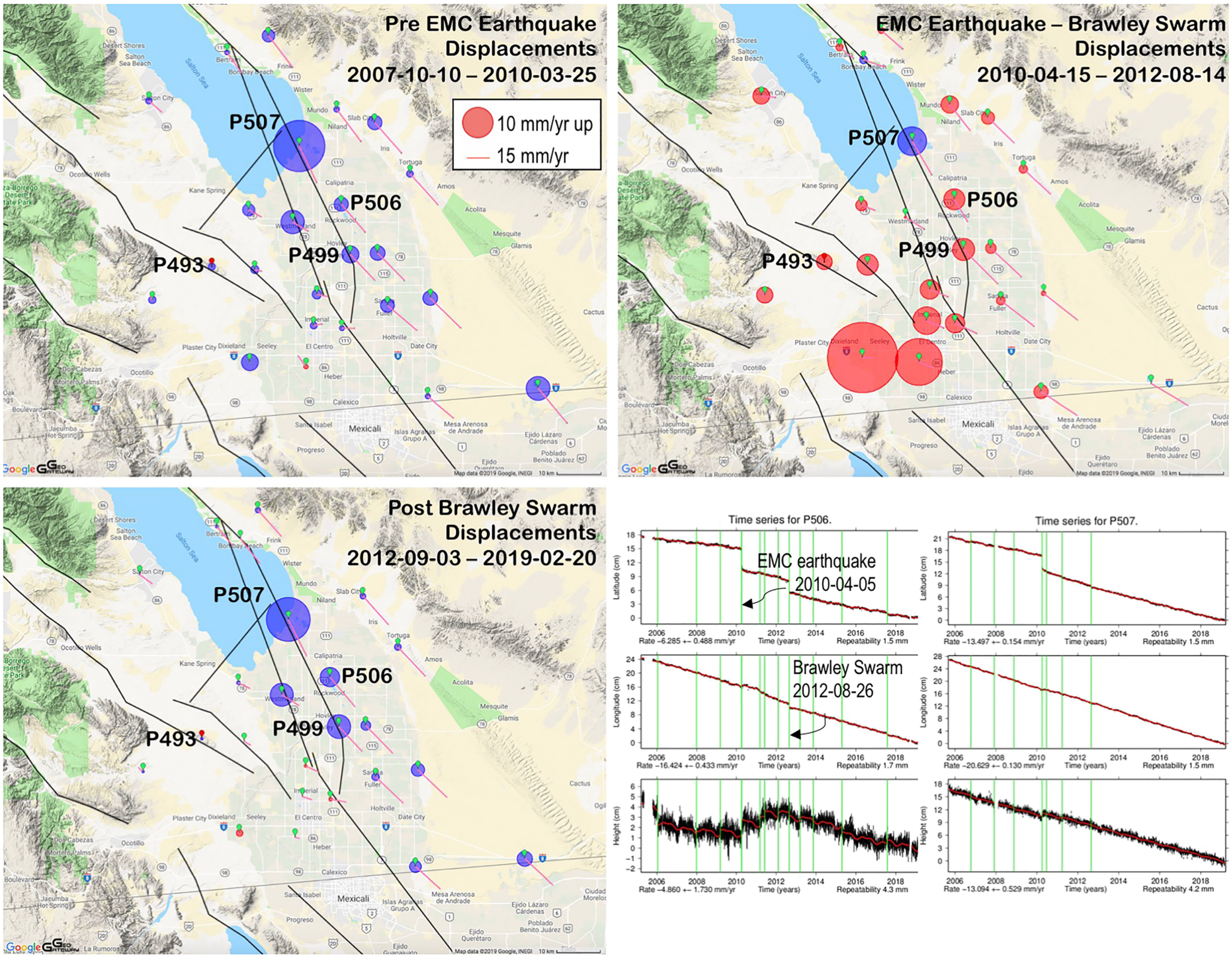

Figure 4. Displacements for three time periods: before the EMC earthquake (top left), between the EMC earthquake and Brawley swarm (top right), and after the Brawley swarm (bottom left). Horizontal velocities are relative to station P493. Vertical velocities in IGS14 are shown. Terrain is shown in the maps, highlighting the Salton Trough. Time series for station P506 also near Brawley like station P499 in Figure 3 shows uplift between the EMC earthquake and Brawley swarm, but subsidence otherwise. Time series for station P507 at the south shore of the Salton Sea shows stable motion, including stable rapid subsidence of 13.1 ± 0.5 mm/year. Displacements have been scaled to mm/year over the given time period for comparison with velocities in Figure 2.

Time series for two stations near Brawley, P499 (Figure 3) and P506 (Figure 4), show a long-term downward motion. Uplift of station P499 and minor subsidence of station P506 at the Brawley swarm terminate a trend of uplift that began after the EMC earthquake. Subsidence for both stations occurs both before the EMC earthquake and after the Brawley swarm. Before the EMC earthquake and after the Brawley swarm, the Salton Trough shows a regional pattern of subsidence. Between these two events the Salton Trough shows a broad pattern of uplift (Figure 4). Station P507 at the southern shore of the Salton Sea is an exception showing a very stable long-term subsidence rate of 13.1 ± 0.5 mm/year. Aside from a southward jump due to the EMC earthquake, the station shows stable rates in all components of motion.

5. Discussion

We determined an initial interpretation of crustal deformation and rate estimates by calculating the velocities relative to station P493, downloading the table produced as output, and then calculating the magnitude

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Table 3
Break Values Shown in Figure 3

Station	Longitude	Latitude	E (mm)	N (mm)	V (mm)
Breaks 26 August 2012					
CRRS	−115.735044	33.069808	−6.9 ± 1.2	0.7 ± 1.4	0.0 ± 5.0
P495	−115.628395	33.044961	−27.2 ± 1.2	4.3 ± 1.4	1.8 ± 4.7
P497	−115.577033	32.834735	2.3 ± 1.2	6.0 ± 1.4	−1.1 ± 4.8
P498	−115.569568	32.898431	3.2 ± 1.2	17.5 ± 1.4	−3.2 ± 5.1
P499	−115.487914	32.979606	48.6 ± 1.0	−22.0 ± 1.2	12.8 ± 4.2
P501	−115.397913	32.875756	5.5 ± 1.2	−3.4 ± 1.4	0.7 ± 4.8
P502	−115.421913	32.982451	20.0 ± 1.1	−6.6 ± 1.4	2.2 ± 4.8
P503	−115.720184	32.948915	−5.9 ± 0.9	−2.5 ± 1.0	3.4 ± 3.5
P506	−115.510199	33.08143	−9.8 ± 1.3	−27.7 ± 1.5	−3.2 ± 5.2
P507	−115.612407	33.199975	0.0 ± 0.8	−4.4 ± 0.9	−0.5 ± 3.2
P509	−115.293919	32.890662	4.3 ± 1.1	−2 ± 1.3	−0.9 ± 4.5
Breaks 7 September 2017					
P503	−115.720184	32.948915	4.8 ± 0.9	−2.5 ± 1.0	−0.8 ± 3.5

of the velocity vector for stations in the table relative to P493. Plotting velocities or other crustal deformation vectors relative to a station can highlight crustal deformation features, focussing attention for further analysis. Plotting IGS14 velocities suggests a clockwise rotation of the Salton Trough with a transfer of southwest North American Plate motion peeling off into northwest motion of the Pacific plate along the western half of the Salton Trough. Plotting velocities relative to station P493 just west of the Salton Trough and near the southern San Jacinto fault suggest that the Imperial fault divides the Pacific and North American plates along a fairly localized zone with block-like motion occurring to the east and west. Lyons et al. (2002) resolved 9 mm/year of right-lateral fault slip at the Imperial fault between 1993 and 2000 and block-like motion on either side of the fault at a rate of 15 mm/year. Our results suggest a qualitative rate difference between GNSS stations of 21 mm/year across the fault in the time period 2005–2019 and a farther field rate of 26 mm/year. Whether the rate has increased since 2005 or GNSS measurements have become more thorough should be the subject of future work. Calculating displacements or velocities relative to stations within the network allows for a initial interpretation of crustal deformation from which more sophisticated models can be developed.

The pattern of deformation becomes less localized along a single fault in the northern half of the Salton Trough, consistent with Meade and Hager (2005). Consistent with other analysis (Lindsey & Fialko, 2013; Meade & Hager, 2005; Thatcher et al., 2016), we find about 11 mm/year of slip across the San Jacinto fault system and about 17 mm/year between the western shore of the Salton Sea and the east side of the southern San Andreas fault. GeoGateway tools can be used to rapidly assess rates of deformation. Extracting velocities relative to station ERRG just north of the Elmore Ranch fault shows a left-lateral relative motion of 4.1 ± 0.3 mm/year and negligible vertical motion for site CRRS about equidistant on the other side of the fault (Figure 2). Future deformation models should take into account active left-lateral faulting along the Elmore Ranch fault in the transtensional Salton Trough. Conducting time-dependent analysis of the GNSS data using these or other tools will help answer whether the plate boundary is shifting westward or bifurcating.

Perhaps the most significant finding of this rapid analysis is the reversal of long-term subsidence of the Salton Trough to sustained uplift for over 2 years following the 2010 EMC earthquake. The steady uplift was abruptly reversed at the 2012 Brawley swarm. Perhaps the swarm released pressure in the Salton Trough crust that had built up as a result of coseismic deformation and postseismic response from the EMC earthquake. Postseismic motion of the Salton Trough computed for the period between the EMC earthquake and Brawley swarm indicates an even more pronounced pattern of uplift, because the postseismic motion reflects residual motions from the long-term rates. In general, however, we find interpretation of vertical motions to be more realistic in IGS14 rather than relative rates or displacements. The pattern of uplift produced as a direct result of the Brawley swarm suggests a graben-like structure or rhombochasm bounded by northeast/southwest striking margins.

The Brawley swarm caused westward motion at the east side of the Superstition Hills fault and no coseismic motion further west. This suggests that the fault clamped as a result of the swarm. The displacement observed on 8 September 2017 is coincident with the M8.2 Chiapas earthquake, indicating that the Chiapas earthquake (Ye et al., 2017), 2,900 km away, triggered slip on the SHF. The creep event accommodated just over 1 year of long-term slip on the fault. The Chiapas earthquake also triggered slip on the Southern San Andreas fault (Tymofyeyeva et al., 2018).

6. Conclusions

GeoGateway tools allow rapid analysis and visualization of GNSS time series and derived crustal deformation and show a complex pattern of behavior of the Salton Trough area in southern California. Horizontal velocities plotted relative to station P493 near the southern San Jacinto fault suggest that the Imperial fault divides the Pacific and North American plates but that the plate boundary bifurcates further north along the eastern and western shore of the Salton Sea, corresponding to the San Andreas and San Jacinto faults, respectively. The Salton Trough broadly marks the boundary between the Pacific and North American plates and is subsiding long term. Uplift occurred in the Salton Trough following the 2010 EMC earthquake and continued until the 2012 Brawley swarm, when the pattern reversed and the trough again entered a phase of long-term subsidence. The 2017 M8.2 Chiapas earthquake triggered right-lateral slip on the Superstition Hills fault.

Data Availability Statement

Readers can access data online (https://sideshow.jpl.nasa.gov/post/series.html).

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